



A rare case of respiratory failure associated with spinal cord injury caused by stabbing with scissors and a knife

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Abstract

In forensic practice, spinal cord injury (SCI) resulting in death has rarely been reported. Here, we present the case of a 65-year-old woman who on admission was conscious without dyspnea or dysphagia. Physical examination revealed two sharp objects penetrating the neck: a pair of scissors lodged in the neck on the right side of the thyroid cartilage and a knife embedded in the nuchal region accompanied by minor seepage of bloody exudate. Radiography showed that the scissors and knife were retained in the cervical spine. Despite a series of medical interventions, the patient died 26 days later. Method: A systematic forensic autopsy was performed. Results: The cause of death was confirmed to be respiratory failure associated with SCI, which was caused by the combination of scissors and a knife. Conclusion: Based on this case, we believe that when there are multiple causes of death, forensic pathologists should determine the primary, immediate, contributory, and other causes of death to ascertain criminal responsibility.

Keywords Forensic pathology · Respiratory failure · Spinal cord injury · Stab wounds

Introduction

Spinal cord injury (SCI) is defined as damage to the intraspinal nerve structure, including the spinal cord and nerve root. Its function is affected by various factors, resulting in dysfunction of the spinal cord at and below the site of the injury [1]. The annual incidence rate of SCI ranges from 11 to 50

cases per million individuals [2, 3], with a higher prevalence in individuals under the age of 30 years [4]. There are several causes of SCI, including motor vehicle accidents, falls, gunshot wounds, and stabbing.

SCIs are associated with a high mortality rate, with estimates of acute in-hospital mortality ranging from 4 to 17% [5]. Here, we present the case of a 65-year-old woman who was stabbed in the neck by a combination of scissors and a knife; after a series of medical interventions, the woman died 26 days later. Through detailed forensic examinations, we determined the cause of death and provided court evidence for the division of criminal responsibility.

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Case report

Clinical summary

A 65-year-old woman was maliciously stabbed in the nuchal region with a knife and in the neck with scissors while fighting with others and was immediately brought to the emergency department of a local hospital. On admission, she was conscious without dyspnea or dysphagia. Physical examination revealed two sharp objects penetrating the neck: a pair of scissors lodged in the neck on the right side of the

thyroid cartilage and a knife embedded in the nuchal region accompanied by minor seepage of bloody exudate. Radiography showed that the scissors and knife were retained in the cervical spine (Fig. 1a, b).

Eight hours after the trauma, medical professionals removed the scissors and knife (Fig. 1c, d). While

operating on the anterior neck, it was evident that the pair of scissors was inserted through the right side of the thyroid cartilage, damaging the upper pole of the thyroid, and stopping at the cervical vertebral plate, with no significant macrovascular injury. In addition, behind the

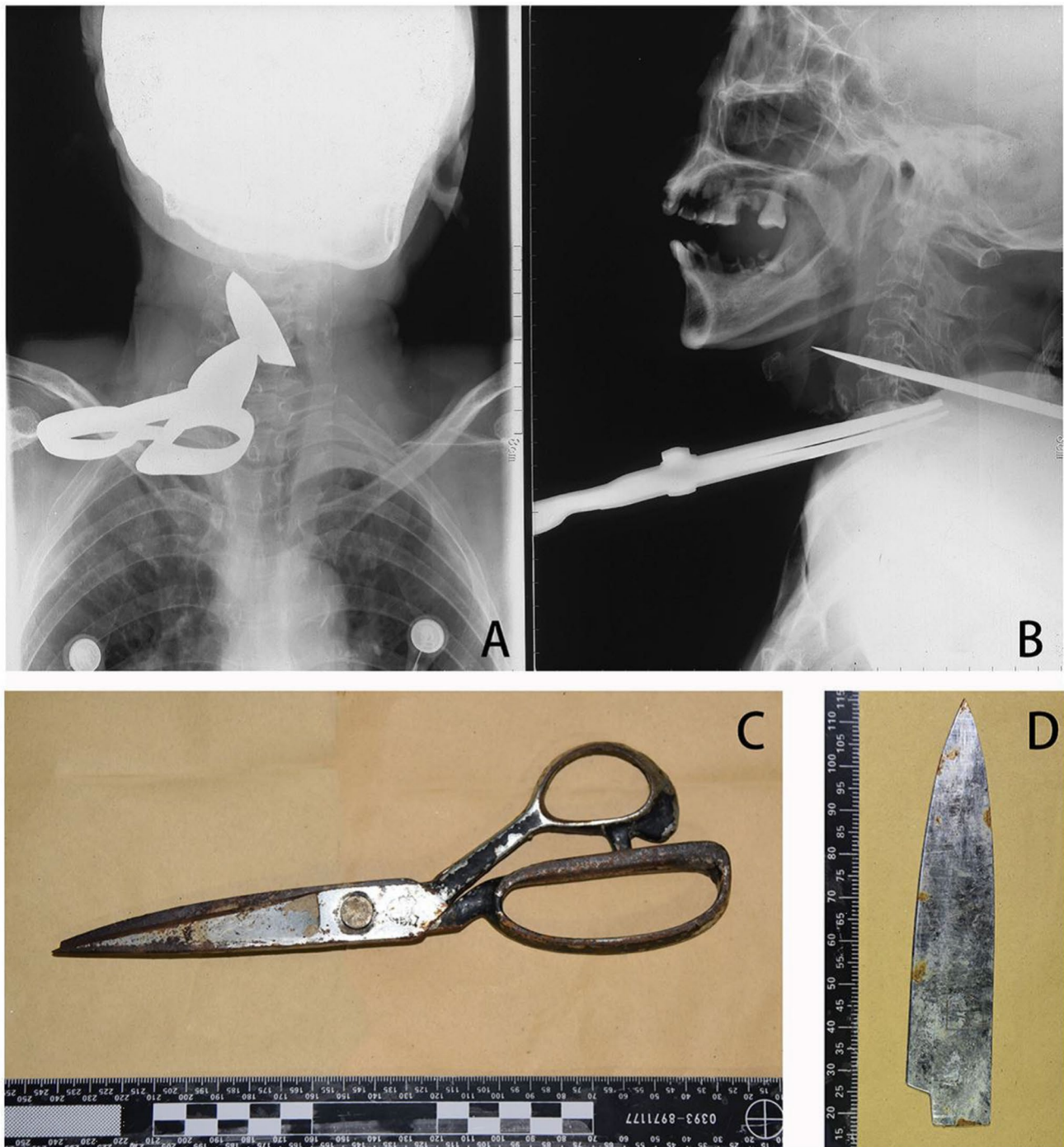


Fig. 1 The cervical vertebrae X-ray scan of the frontal position (a) and the right lateral position (b). The scissors (c) and the knife blade (d) that caused the damage are below the image

fourth cervical vertebra, a 2.0 cm long knife handle was observed, and light red bloody fluid flowed from the point of penetration.

Seven days later, she experienced chest tightness and shortness of breath. On examination, blood oxygen saturation was only 75%, and wet rales were heard in both lungs. In addition, atelectasis of the lower left lung and a small amount of bilateral pleural effusion were observed on computerized tomography (CT), suggesting the presence of pulmonary inflammation. Bacterial cultures of the woman's sputum revealed multidrug-resistant *Stenotrophomonas maltophilia* and gram-positive cocci. Thirteen days later, the patient experienced intermittent fever, poor cough reflex, T: 37.3°, P: 72 times/minute, R: 20 times/minute (ventilator-assisted breathing), blood pressure: 125/78 mm Hg, and shortness of breath; and wet rales were heard in both lungs. CT examination revealed stripes and patchy high-density shadows in the upper and middle lobes of both lungs (Fig. 2). Subsequently, type II respiratory failure and pulmonary infection exacerbated the disease, leading to her death on the 26th day after the trauma.

Forensic autopsy

Owing to the homicidal nature of death, a comprehensive autopsy was performed to determine the cause of death. The external examination revealed a 6.0 cm × 1.0 cm sutured skin wound in the neck region located on the right side of the thyroid cartilage, and a sutured skin wound (1.5 cm × 1.5 cm) located in the nuchal region. Multiple abrasions and contusions were observed on the head and other body surfaces.

The left and right lungs weighed 500 g and 950 g, respectively, and a 5.0 cm long thrombus-like clot was observed

in the right pulmonary artery. Microscopic examination revealed that the clot was composed of platelet beams, red blood cells, neutrophils, protein exudates, and fibrin (Fig. 3). Thromboembolisms or microthrombi were observed in the small blood vessels. Multiple foci and flaky hemorrhages in the alveoli, neutrophil inflammatory infiltration in the alveoli and bronchioles, and small abscesses were also observed.

The fracture was located on the right side of the fifth anterior cervical vertebral body (Fig. 4). A small hematoma measuring 1.0 cm × 0.5 cm was located at the submucosal at the edge of the right thyroid cartilage plate, and a gray-white granulation tissue scar formation was seen on section. The thyroid capsule was incomplete, reduced, and contained atrophied follicles; new granulation tissue and hemorrhages were also observed. Lymphocytes, plasma cells, fibroblasts, multinucleated foreign-body giant cells, and new capillaries formed the granulation tissue.

The cervical spinal cord was swollen in appearance, and the section showed local tissue necrosis and softening with a grayish-white appearance in the dorsal mid-right area of the cervical spinal cord (Fig. 5a). A thickened arachnoid, a variable number of patchy hemorrhages, mononuclear and lymphocyte infiltration, and hemosiderin granules in macrophages were seen. Neuronal necrosis, demyelinated axons, foam cells, gliosis, scattered areas of degenerative changes with loss of the usual fibrillary pattern, and decreased staining affinity were observed (Fig. 5b). The same scattered areas of degeneration were also observed with Luxol fast blue myelin staining (Fig. 5c). Microscopic examinations of the heart and kidneys revealed no histopathological changes. No conventional toxins were detected in the stomach.

Considering the circumstances of this case, the patient's medical records, and the postmortem pathological changes,

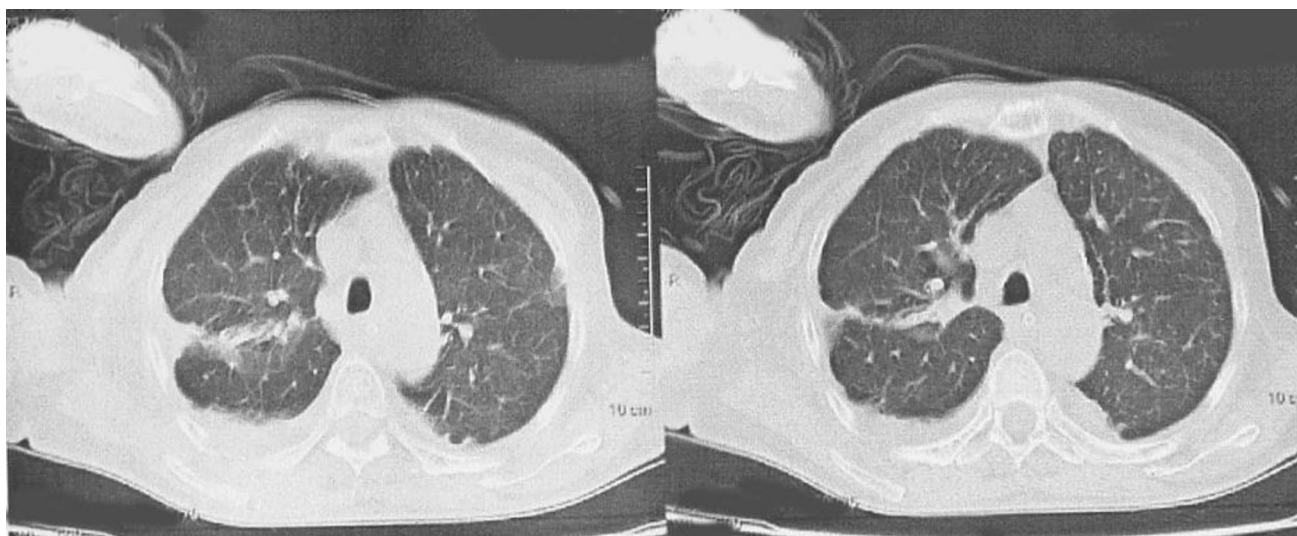


Fig. 2 CT examination shows plaque-shaped high-density shadows in the upper lobe of both lungs

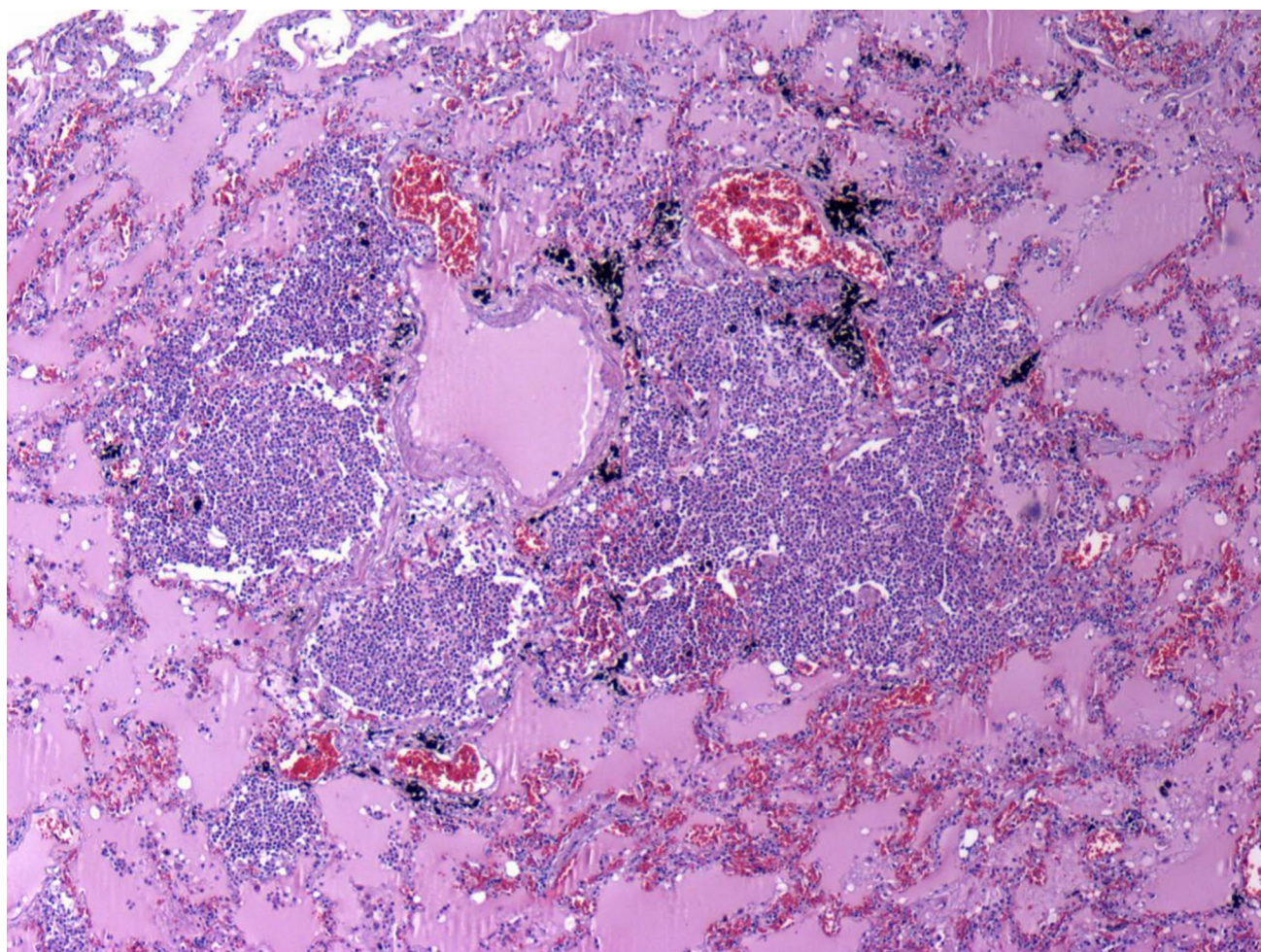


Fig. 3 Microscopy of a thrombus-like clot in the right pulmonary artery shows that the clot was composed of platelet beams, red blood cells, neutrophils, protein exudates, and fibrin. Hematoxylin and eosin 1:100

Fig. 4 The fracture was located on the right side of the fifth anterior cervical vertebral body



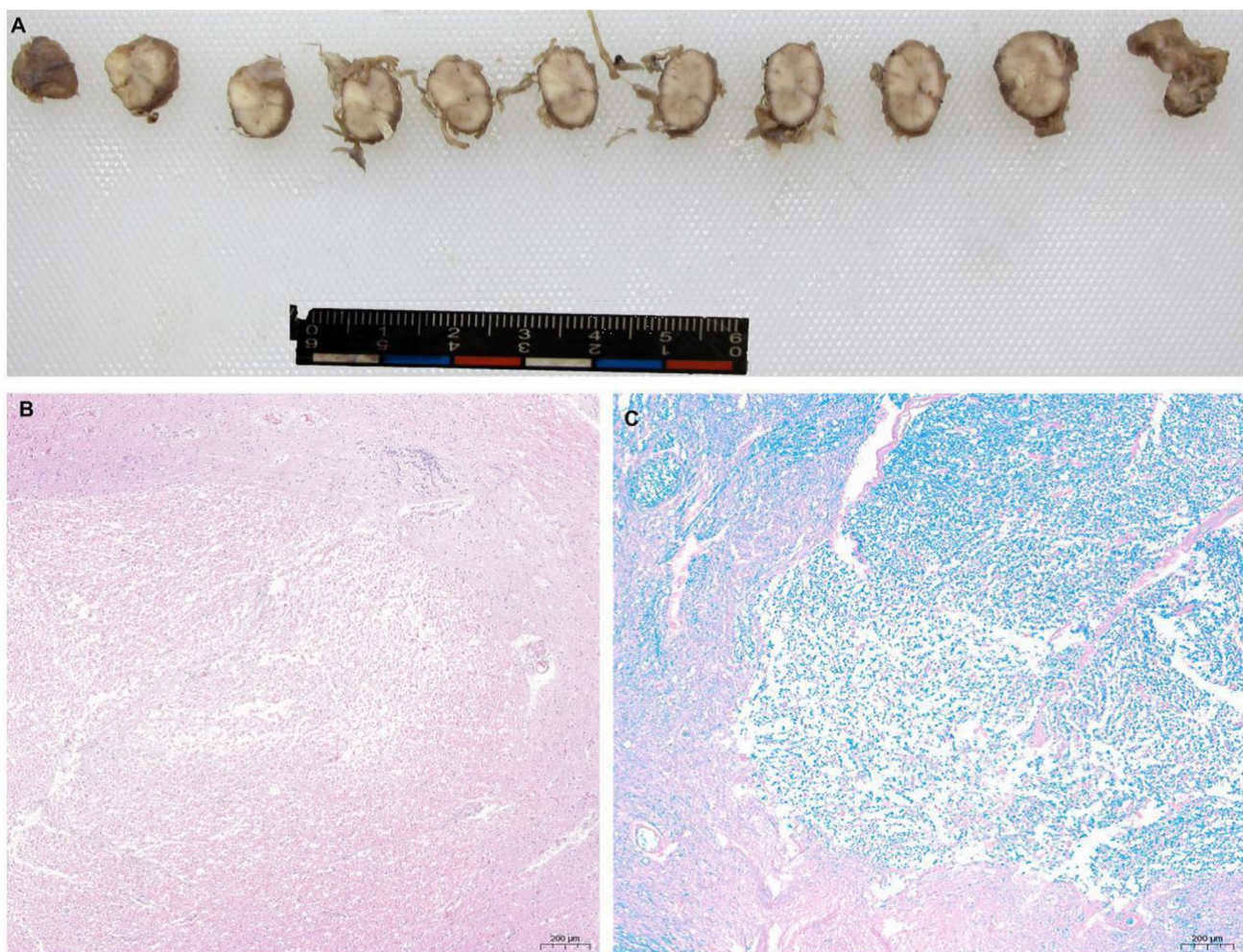


Fig. 5 Spinal cord (a) and transverse sections of the cervical marrow stained with hematoxylin and eosin (b) and Luxol fast blue (c). a: section showing local tissue necrosis and softening with a grayish-

white appearance in the dorsal mid-right area of the cervical spinal cord. b, c: scattered areas of degeneration. Scale bars: b, c = 200 μ m

we excluded mechanical asphyxia and poisoning, and determined that the cause of death was respiratory failure caused by SCI, which was the result of stabbing with the combination of scissors and a knife.

Discussion

Stab wounds in the spinal cord have rarely been reported [6]. Peacock et al. reviewed 1600 patients with SCI and found that 25% of cases were caused by sharp injuries, and the most common site of stab wounds was the thoracic (63.8%), followed by the cervical (29.6%), and lumbar vertebrae (6.7%) [7]. Owing to the protective effect of the cervical vertebrae body and spinous on the spinal cord, the puncture direction and position of sharp instruments can result in different consequences. In this case, the vertebral body stopped the scissors stabbed from the anterior region, whereas the

knife blade stabbed from behind injured the spinal dura and spinal cord, resulting in SCI.

The in-hospital mortality rate of acute traumatic SCI ranges from 4 to 17% [5]. The risk of mortality increases with the level of injury, implying a higher mortality rate in patients with cervical SCI than in those with lumbar SCI. Patients with cervical SCI are prone to complications, and respiratory complications are among the most serious and common, often associated with death. Injuries above C5 disrupt diaphragm innervation, leading to compromised respiratory function. In this case, because of the SCI, the patient was bedridden and on a ventilator, resulting in a nosocomial infection, which was characterized by multidrug-resistant *S. maltophilia* and gram-positive cocci. The infection ultimately led to respiratory failure, resulting in death.

In forensic pathology, SCIs caused by mechanical injuries usually manifest as hemorrhage, necrosis, and inflammation [5, 8–13]. After surviving the first 72 h after injury, patients

with cervical SCI show a higher incidence of sepsis and multiple organ failure [14]. People with SCI may experience various delayed medical complications of paralysis that can lead to death months to years after injury. In this case, we observed multiple lesions with patchy hemorrhages in the alveoli, neutrophilic inflammatory infiltrates in the alveoli and bronchi, and the formation of small abscesses. These changes were indicative of an infection that occurred during treatment and led to death from respiratory failure. In addition, the death of this patient occurred 26 days after the spinal cord injury, which is consistent with the above-mentioned post-injury complications.

Because the deceased was attacked and subsequently infected in the hospital, it is necessary to allocate responsibility in the case, which requires a forensic pathologist to clearly determine the cause of death, including the underlying cause of death, immediate cause of death, and other causes of death. In this case, based on the available case information, medical records, and forensic examinations, we determined that the deceased suffered multiple sharp force injuries to the neck and cervical medullary regions and later died of respiratory failure due to infection. Respiratory failure was determined to be the immediate cause of death, and the scissors and knife stab wounds to the neck were the underlying causes.

In summary, we report a rare case of cervical SCI due to the combined action of scissors and a knife on the cervical collar, which resulted in death due to systemic respiratory failure from pulmonary infection during hospitalization. In such cases, forensic pathologists should be clear about their duties, treat each case seriously, clarify the cause of death, and improve the forensic expert opinion's evidentiary power in court.

Key points

1. We report a case of a 65-year-old woman with a spinal cord injury (SCI), which was caused by being stabbed in the neck with scissors and a knife.
2. The immediate cause of death was confirmed to be respiratory failure.
3. The SCI caused by the stab wounds to the neck was the underlying cause.
4. Forensic pathologists should build convincing links to the original SCI to medical complications to illustrate the manner of death.

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Declarations

Ethical approval The case was approved by the ethics committee of Huazhong University of Science and Technology.

Informed consent Written informed consent was obtained from the family members of the patient.

Conflict of interest The authors declare that they have no conflict of interest.

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